AI DUO

AN INTERNATIONAL AWARD-WINNING INSTITUTION FOR SUSTAINABILITY

AI MATHEMATICAL OLYMPIAD

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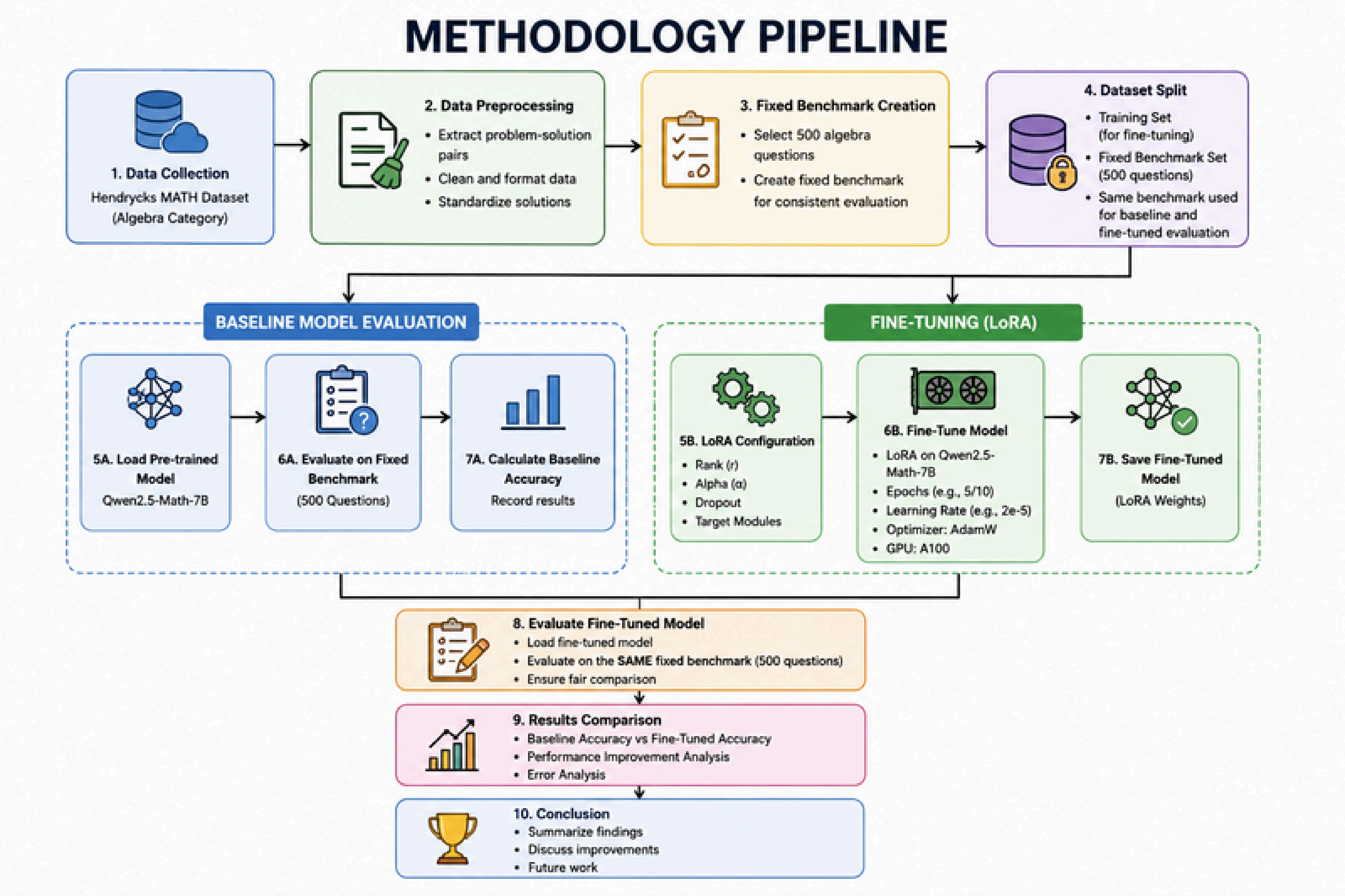
CONTENTS

INTRODUCTION

- Solving Olympiad-level mathematics remains difficult for current language models.
- Designed to evaluate mathematical reasoning rather than simple memorization.
- Fine-tuning provides an opportunity to adapt pretrained models for specialized reasoning tasks.

METHODOLOGY

1. DATASET PREPARATION
2. DATA PREPROCESSING AND TOKENIZATION
3. LOAD PRETRAINED QWEN2.5-MATH-7B
4. APPLY LoRA (LOW-RANK ADAPTION)
5. FINE-TUNE THE MODEL (5 EPOCHS)
6. EVALUATE ON DATASET
7. GENERATE FINAL PREDICTIONS


METHODOLOGY PIPELINE
1. Data Collection
Hendrycks MATH Dataset
(Algebra Category)
2. Data Preprocessing
• Extract problem-solution pairs
• Clean and format data
• Standardize solutions
3. Fixed Benchmark Creation
• Select 500 algebra questions
• Create fixed benchmark for consistent evaluation
4. Dataset Split
• Training Set (for fine-tuning)
• Fixed Benchmark Set (500 questions)
• Same benchmark used for baseline and fine-tuned evaluation
BASELINE MODEL EVALUATION
5A. Load Pre-trained Model
Qwen2.5-Math-7B
6A. Evaluate on Fixed Benchmark
(500 Questions)
7A. Calculate Baseline Accuracy
Record results
FINE-TUNING (LoRA)
5B. LoRA Configuration
• Rank (r)
• Alpha (α)
• Dropout
• Target Modules
6B. Fine-Tune Model
• LoRA on Qwen2.5-Math-7B
• Epochs (e.g., 5/10)
• Learning Rate (e.g., 2e-5)
• Optimizer: AdamW
• GPU: A100
7B. Save Fine-Tuned Model
(LoRA Weights)
8. Evaluate Fine-Tuned Model
• Load fine-tuned model
• Evaluate on the SAME fixed benchmark (500 questions)
• Ensure fair comparison
9. Results Comparison
• Baseline Accuracy vs Fine-Tuned Accuracy
• Performance Improvement Analysis
• Error Analysis
10. Conclusion
• Summarize findings
• Discuss improvements
• Future work

EXPERIMENTAL SETUP

1

BASE MODEL

Qwen2.5-Math-7B (Pretrained Large Language Model)

2

FINE-TUNING METHOD

Low-Rank Adaptation (LoRA)

3

HARDWARE

NVIDIA A100 GPU

PARAMETER	VALUE
Number of Epochs	5
Fine-Tuning Method	LoRA
Maximum Sequence Length	1024 tokens
Maximum Generation Length	512 tokens
Framework	PyTorch + Hugging Face + PEFT

RESULTS

MODEL : QWEN2.5-MATH-1.5B

WITHOUT FINE-TUNING

```
FINAL PERFORMANCE SUMMARY
==========================================================
Dataset Used            : Hendrycks MATH
Total Questions Tested : 500
Correct Answers         : 78
Incorrect Answers       : 422
Final Accuracy          : 15.60%
==========================================================
```

FINE-TUNING

```
FINAL PERFORMANCE SUMMARY
==========================================================
Dataset Used            : Hendrycks MATH
Model Type              : Fine-Tuned Qwen2.5-Math + LoRA
Total Questions Tested : 500
Correct Answers         : 53
Incorrect Answers       : 447
Final Accuracy          : 10.60%
==========================================================
```

RESULTS

MODEL : QWEN2.5-MATH-7B

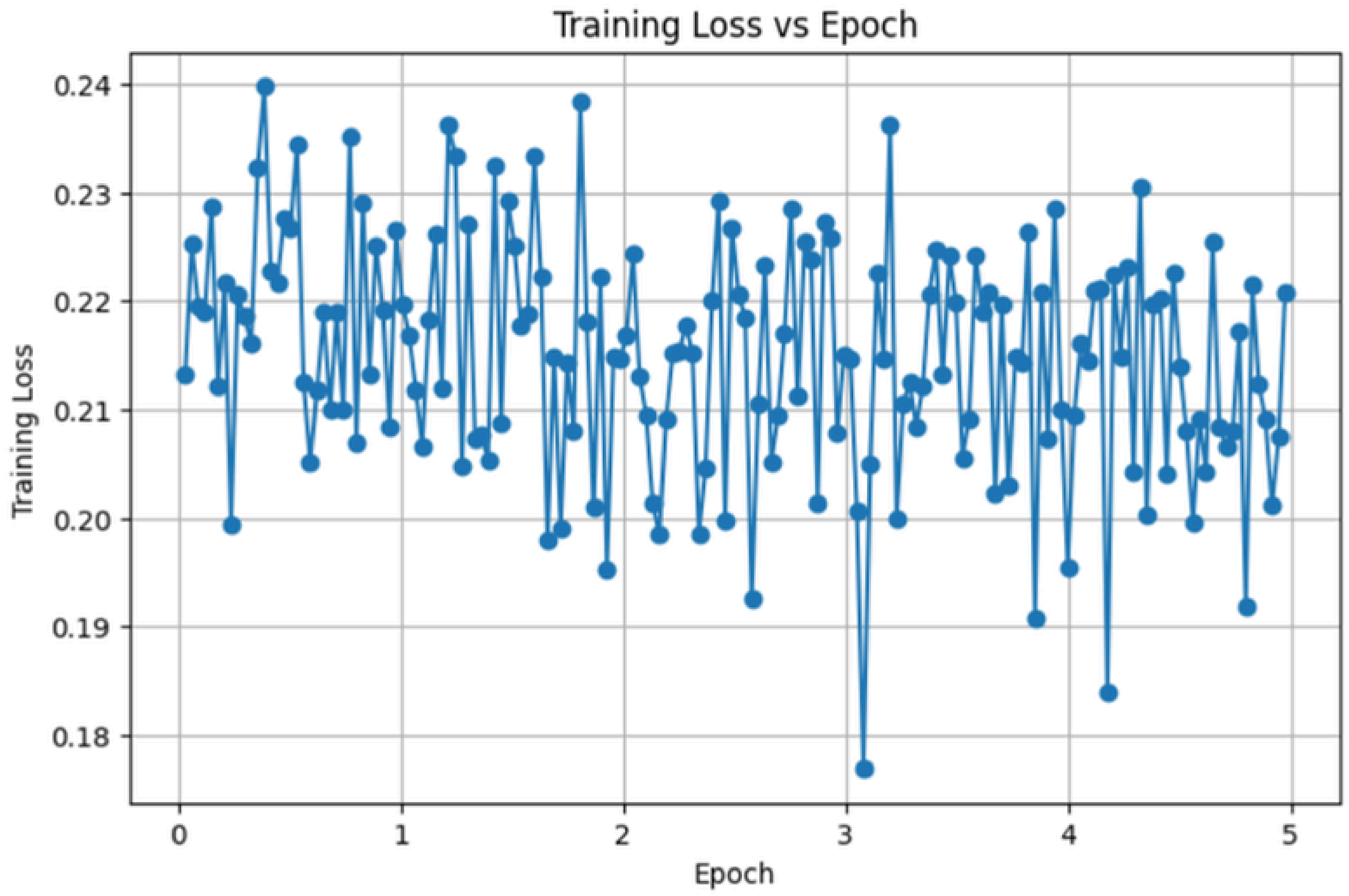
WITHOUT FINE-TUNING

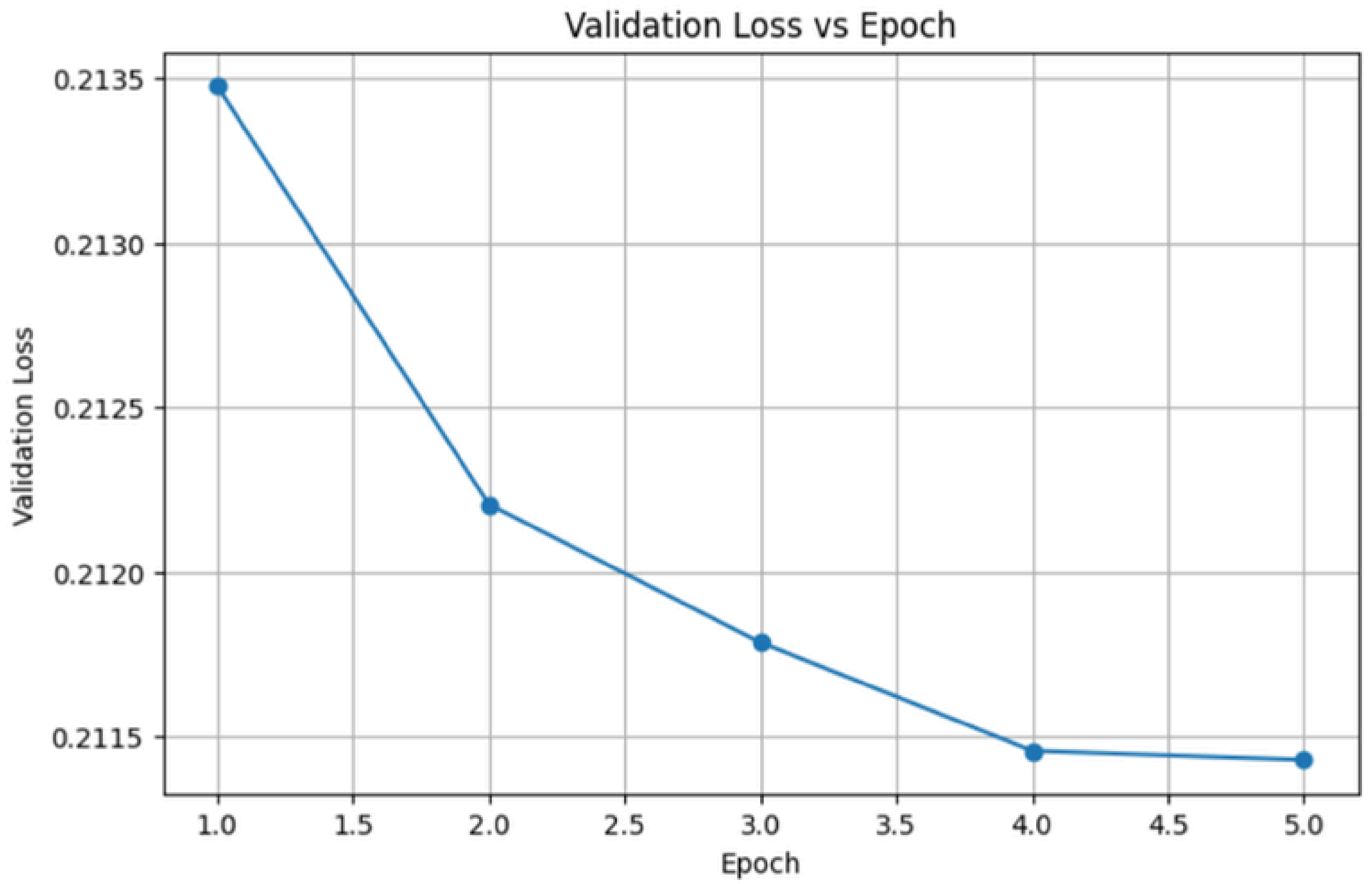
```
FINAL PERFORMANCE SUMMARY
===========================================================
Dataset Used           : Hendrycks MATH
Total Questions Tested : 500
Correct Answers        : 75
Incorrect Answers      : 425
Final Accuracy         : 15.00%
```

FINE-TUNING

```
============================================================
FINAL PERFORMANCE SUMMARY
============================================================
Dataset Used           : Hendrycks
Model                  : Qwen2.5-Math-7B + LoRA
Final Accuracy         : 57.00%
============================================================
```

RESULTS


Training Loss vs Epoch
Training Loss
0.24
0.23
0.22
0.21
0.20
0.19
0.18
0
1
2
3
4
5
Epoch


Validation Loss vs Epoch
Validation Loss
0.2135
0.2130
0.2125
0.2120
0.2115
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Epoch

DISCUSSION

STRENGTHS

- Parameter-efficient adaptation through LoRA.
- Lower computational cost than full fine-tuning.
- Ability to leverage pretrained mathematical knowledge.

LIMITATIONS

- Model performance depends on dataset quality.
- Some outputs may still contain formatting or reasoning errors.
- Exact-match evaluation may not capture mathematically equivalent answers expressed differently.

FUTURE WORKS

- Increase the size and diversity of the training dataset.
- Experiment with different hyperparameter settings such as learning rate and batch size.
- Compare LoRA with other parameter-efficient fine-tuning methods.
- Improve answer extraction and evaluation to recognize mathematically equivalent expressions.

CONCLUSION

- The Qwen2.5-Math-7B model was successfully fine-tuned using LoRA.
- Fine-tuning enabled the pretrained model to better adapt to mathematical reasoning tasks while requiring fewer trainable parameters.
- Training was performed for 5 epochs using an NVIDIA A100 GPU.
- The proposed approach demonstrates that parameter-efficient fine-tuning is a practical method for improving specialized AI models.

Thank You!

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